



Store at: +2+8°C.

Presentation:

Cod. EZ009LQ CONT: R1 1 x 100 + R2 1 x 25 mL.

EZ010LQ CONT: R1 2 x 100 + R2 2 x 25 mL.

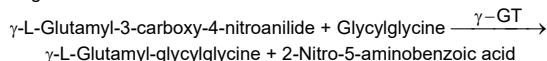
Procedure

Quantitative determination of gamma-glutamyl transferase (γ -GT).

Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

γ - glutamyl transferase (γ -GT) catalyses the transfer of γ - glutamyl group from γ - glutamyl-p-nitroaldehyde to acceptor glycylglycine according to the following reaction:



The rate of 2-Nitro-5-aminobenzoic acid formation, measured photometrically, is proportional to the catalytic concentration of γ - GT present in the sample^{1,2}.

REAGENTS COMPOSITION

R 1 Buffer	TRIS Glycylglycine	100 mmol/L. 100 mmol/L.
R 2 Substrate	L- γ -glutamyl-3-carboxy-4-nitroanilide	3 mmol/L.

REAGENT PREPARATION AND STABILITY

Working reagent (WR):

Mix: 4 vol. (R1) Buffer + 1 vol. (R2) Substrate

Stability: 21 days at 2-8°C or 5 days at room temperature (15-25°C).

Signs of Reagent deterioration:

- Presence of particles and turbidity.
- Blank absorbance (A) at 405 nm. ≥ 1.20

All the components of the kit are stable until the expiration date on the label when stored at 2-8°C, protected from light and contamination prevented during their use. Do not use reagents over the expiration date.

SPECIMEN

Serum¹. Gamma-GT Stability: 3 days at 2-8°C. 8 hours at 15-25°C and 1 month at -20°C.

MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 405 nm.
- Thermostatic bath at 25°C, 30°C or 37°C ($\pm 0.1^\circ\text{C}$).
- Matched cuvettes 1.0 cm. light path.

General laboratory equipment.

TEST PROCEDURE

- Assay Conditions
 - Wavelength: 405 nm.
 - Cuvette: 1 cm light path.
 - Constant temperature 25°C / 30°C / 37°C.
- Adjust the instrument to zero with distilled water or air.
- Pipette into a cuvette (note 1):

WR (mL.)	1.0
Sample (μL .)	100

- Mix and incubate for 1 minute.
- Read the absorbance (A) of the sample, start the stopwatch and read absorbance at 1 min. interval thereafter for 3 min.
- Calculate the difference of absorbance and the average absorbance difference per minute ($\Delta A/\text{min}$.)

CALCULATIONS

$$\Delta A/\text{min} \times 1190 = \text{U/L of } \gamma\text{-GT (note 2)}$$

Units: One international unit (IU) is the amount of enzyme that transforms 1 μmol of substrate per minute, in standard conditions. The concentration is expressed in units per liter of sample (U/L).

Temperature conversion factors

To correct results to other temperatures multiply by:

Assay temperature	Conversion factor to		
	25°C	30°C	37°C
25°C	1.00	1.37	1.79
30°C	0.73	1.00	1.30
37°C	0.56	0.77	1.00

QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, Control H Normal Ref. QC003 and Control H Pathological Ref. QC004. If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions

REFERENCE VALUES¹

	25°C	30°C	37°C
Women	4 - 18 U/L.	5 - 25 U/L.	7 - 32 U/L.
Men	6 - 28 U/L.	8 - 38 U/L.	11 - 50 U/L.

(Estos valores son orientativos).

It is suggested that each laboratory establish its own reference range

CLINICAL SIGNIFICANCE

γ - GT is a cellular enzyme with wide tissue distribution in the body, primarily in the kidney, pancreas, liver and prostate.

Measurements of γ - GT, activity are used in the diagnosis and treatment of hepatobiliary diseases such biliary obstruction, cirrhosis or liver tumours^{1,2,5,6}.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

Measuring Range:

From detection limit of 2 U/L. to linearity limit of 300 U/L., under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample 1/2 with NaCl 9 g/L. and multiply result by 2.

Precisión:

	Intra-assay n= 20		Inter-assay n= 20	
Mean (U/L)	38.3	190	40.1	198
SD	0.39	0.53	0.82	2.30
CV (%)	1.03	0.28	2.05	1.16

Sensitivity: 1 U/L = 0.0008 $\Delta A/\text{min}$

Accuracy: Results obtained GPL reagents did not show systematic differences when compared with other commercial reagents.

The results obtained using 100 samples were the following:

Correlation coefficient (r^2): 0.99990

Regression Equation: $y = 1.334x - 1.493$

The results of the performance characteristics depend on the analyzer used.

INTERFERING SUBSTANCES

- Plasma should not be used anticoagulants inhibit the enzyme. Hemolysis interferes with the assay¹
- A list of drugs and other interfering substances with γ -GT determination has been reported^{3,4}.

NOTES

- Use clean disposable pipette tips for its dispensation.
- Formulation to reach constant:

$$\Delta A/\text{min} \times 1190 = \text{U/L of } \gamma\text{-GT}$$

$$\frac{\Delta A/\text{min} \times 1000}{\epsilon \times \text{LP} \times \text{Sv}} = \frac{\text{TV} \times 1000}{\epsilon \times \text{LP} \times \text{Sv}}$$

TV= Total volume in mL
 ϵ 2-nitro-5-aminobenzoic acid = 9.9 at 405 nm
 LP= Light path
 Sv= Sample volume in mL

BIBLIOGRAPHY

- Gendler S. γ -GT. Kaplan A et al. Clin Chem The C.V. Mosby Co. St Louis. Toronto. Princeton 1984; 1120-1123.
- Persijn J P et al. J Clin Chem Clin Biochem 1976; (14) 9: 421-427.
- Young DS. Effects of drugs on Clinical Lab. Tests, 4th ed AACC Press, 1995.
- Young DS. Effects of disease on Clinical Lab. Tests, 4th ed AACC 2001.
- Burtis A et al. Tietz Textbook of Clinical Chemistry, 3rd ed AACC 1999.
- Tietz N W et al. Clinical Guide to Laboratory Tests, 3rd ed AACC 1995.



CHEMELEX, S.A.
 Pol. Ind. Can Castells. C / Industria 113, Nave J
 08420 Canovelles -BARCELONA-
 Tel- 34 93 849 17 35 Fax- 34 93 846 78 75

CHEMELEX, S.A.

Pol. Ind. Can Castells
 C/ Industria 113, Nave J
 08420 - CANOVELLES (Barcelona)
 Tel: 34 93 849 17 35
 Fax: 34 93 846 78 75

GPLBEDTT29-I
 Rev.7 - 03/02/2020